HERIOT WATT UNIVERSITY

B37VR ROBOTICS PROJECT

1 Introduction to Arduinos and using them for reading sensor values

January '21

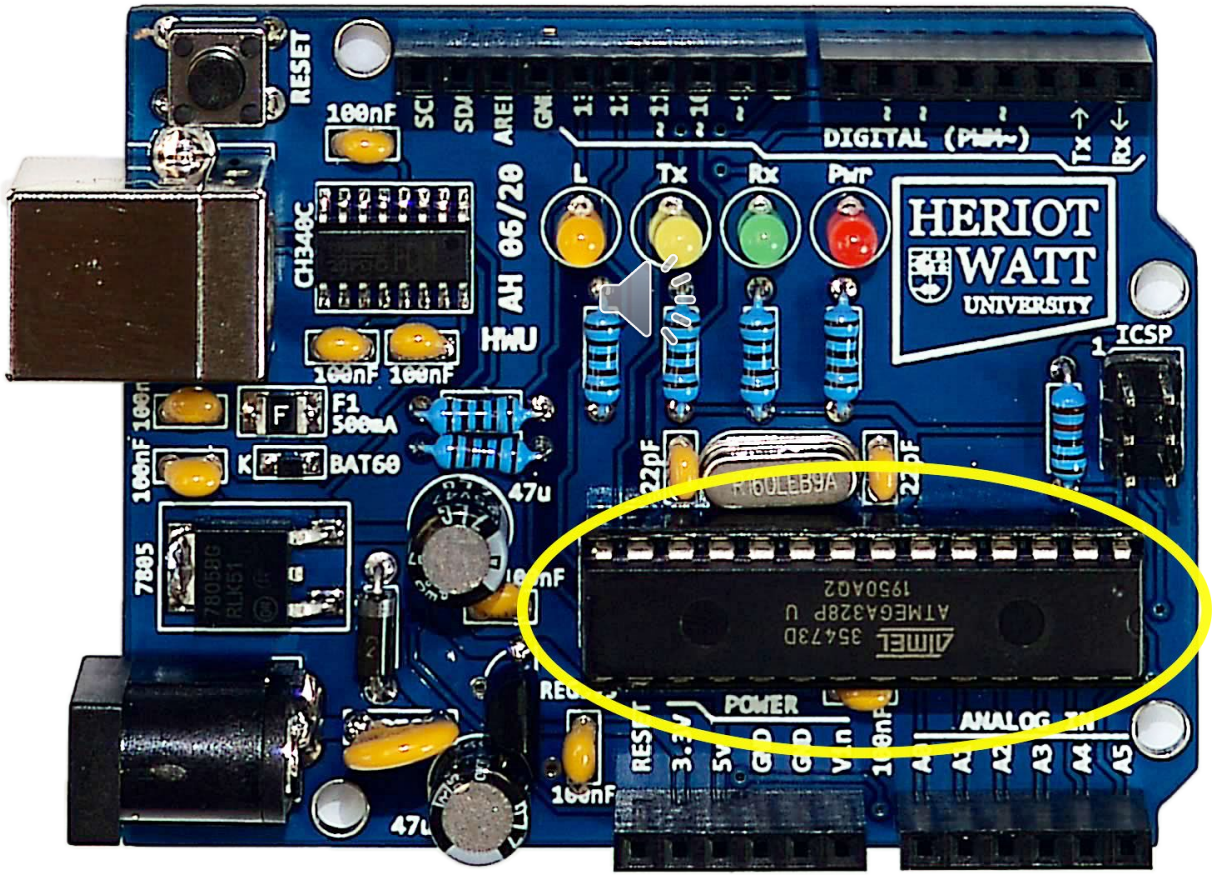
1 SUMMARY: INTRODUCTION TO ARDUINOS AND USING THEM FOR READING SENSOR VALUES

- **LEARNING RESOURCES AVAILABLE**
  - Arduino Arduino IDE
  - Arduino tutorial
  - Light Dependant Resistors (LDRs)
  - Resistors and connection wires
- **LEARNING ACTIVITIES**
  - Look at Arduino tutorial
  - Look at working program
  - Show operation of LDRs in voltage divider circuits
  - Discuss what is needed in characterisation experiments
  - Discuss what is to be expected in laboratory reports

TASKS TO BE UNDERTAKEN

- **TASKS TO BE COMPLETED**
  - Install Arduino IDE on computer you are going to use
  - Run through basic Arduino getting started tutorial
  - Program blink and any other programs to show you can run Arduino programs
  - Set up LDR with resistor as a voltage divider
  - Measure and record the output of the voltage divider using Arduino
  - Set up experiment and take measurements to characterise the LDR

ARDUINO BOARD



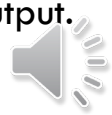
RUNNING A PROGRAM ON THE ARDUINO

- **First you should install the Arduino IDE on the computer you are using.**
- **Installing your first program**
- Now you are ready to test your Arduino by installing a small test program.
- Open the Arduino IDE.
- Under the Tools tab check that **Board: "Arduino/Genuino Uno"** is selected.
- Under the Tools tab set the **PORT: COMxxx** to the same number as when you checked the USB.
- Go to the Files tab and select –
  - Examples
    - 01.Basics
      - **Blink** (left click on this)
- A sketch window called **Blink** will open.
- Left click on the tab with the right pointing arrow to compile and upload the program to your board.
- At the bottom of the screen you will see the message **Compiling sketch** and there will be a green progress bar. If all is OK this message will change to **Uploading…** then **Done uploading**. Whilst uploading you should see activity on the L, Rx and TX LEDs when the program is loading then the L LED will blink at 0.5Hz showing that the program is running.

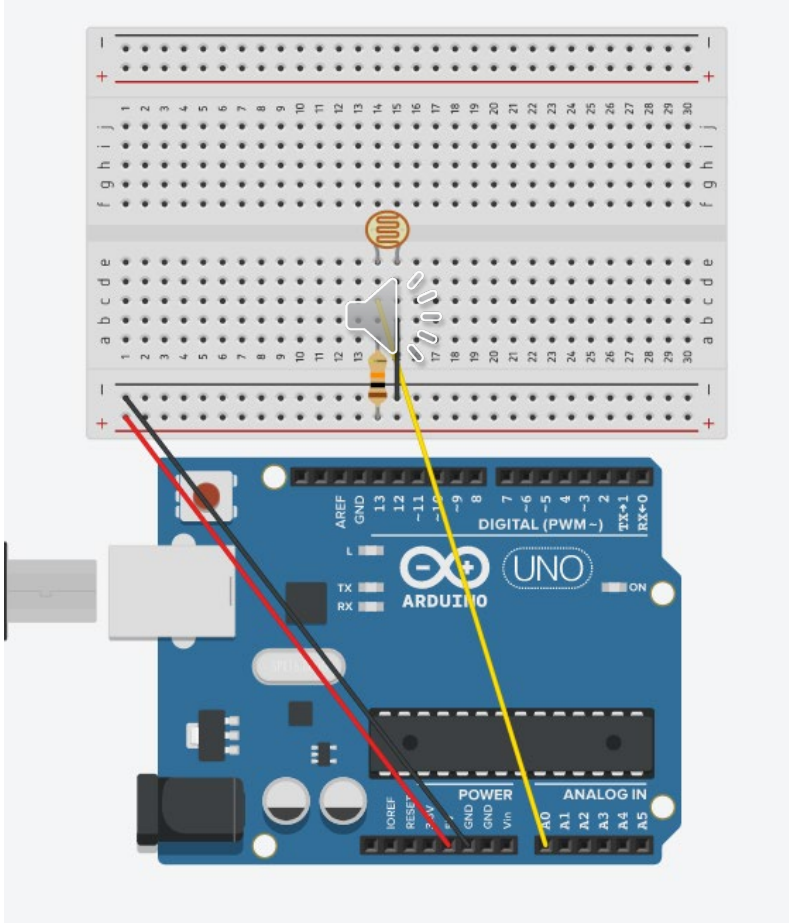
BLINK PROGRAM

Code for Blink This example code is in the public domain
http://www.arduino.cc/en/Tutorial/Blink

```
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}
// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);   // turn the LED on (HIGH is the voltage level)
  delay(1000);                       // wait for a second
  digitalWrite(LED_BUILTIN, LOW);    // turn the LED off by making the voltage LOW
  delay(1000);                       // wait for a second
}
```



CIRCUIT TO READ LDR



SKETCH TO READ OUTPUT OF LDR VOLTAGE DIVIDER CIRCUIT

```
/*
  Based on AnalogReadSerial from Arduino examples
  Reads an analog input on pin 0,
  prints the result to the Serial Monitor.
  Attach the center pin of a voltage divider to pin A0,
  and the outside pins to +5V and ground.
*/
// the setup routine runs once when you press reset:
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
}
// the loop routine runs over and over again forever:
void loop() {
  // read the input on analog pin 0:
  int sensorValue = analogRead(A0);
  // print out the value you read:
  Serial.print("Sensor value =");
  Serial.println(sensorValue);
  delay(1);        // delay in between reads for ADC stability
  delay(5000); // 5s delay to allow results to be read adjust
               //for faster or slower operation
 }
```

WANT TO CHARACTERISE THE LIGHT SENSOR

- Things to look at are the value of the resistor that is part of the voltage divider circuit
- Which way round you want to connect the LDR resistor voltage divider (The LDR at the 0V end or the 5V end)
- What range of values you get for the sensor being completely covered, normal light and bright light
- A comparison of two sensors under the same conditions as it will be based on comparing two sensors that we perform the light following

GUIDANCE ON WHAT SHOULD BE INCLUDED IN A LABORATORY REPORT

Laboratory Report Structure Guidance

The word counts suggested here do not need to be strictly adhered to they are for some basic guidance.



Introduction (approx**:** 150 words**)**

Give a brief description of the aims of the laboratory and a summary of the results collected. Please write this as constant prose in a way that a fellow student, who does not know anything about the experiment, can understand the approach.

Theory and Technical Background (approx: 200 words**)**

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

LABORATORY STRUCTURE CONTINUED

Method (approx: 300 words**)**

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

Results

Add all appropriate figures and analysis. No formal word count for this section, however please keep text as concise as possible.

Conclusions (approx: 100 words**)**

Please give the conclusions from the characterisations including a summary of the results collected.

TASKS TO BE COMPLETED

- Install Arduino IDE on computer you are going to use
- Run through basic Arduino getting started tutorial
- Program blink and any other programs to show you can run Arduino programs
- Set up LDR voltage divider circuit and recording values

